Recursive Training Loops in LLMs: How training data properties modulate distribution shift in generated data?

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Abstract

Large language models (LLMs) are increasingly used in the creation of online content, creating feedback loops as subsequent generations of models will be trained on this synthetic data. Such loops were shown to lead to *distribution shifts* - models misrepresenting the true underlying distributions of human data (also called *model collapse*). However, how human data properties affect such shifts remains poorly understood. In this paper, we provide the first empirical examination of the effect of such properties on the outcome of recursive training. We first confirm that using different human datasets leads to distribution shifts of different magnitudes. Through exhaustive manipulation of dataset properties combined with regression analyses, we then identify a set of properties associated with distribution shift magnitudes. Lexical diversity is found to amplify these shifts, while semantic diversity and data quality mitigate them. Furthermore, we find that these influences are highly modular: data scrapped from a given internet domain has little influence on the content generated for another domain. Finally, experiments on political bias reveal that human data properties affect whether the initial bias will be amplified or reduced. Overall, our results portray a novel view, where different parts of internet may undergo different types of distribution shift.

1 Introduction

Large Language Models (LLMs) are increasingly contributing to the creation of internet content, being used for journalism (Brigham et al., 2024), coding (Jiang et al., 2024) and generating content on social media (Ferrara et al., 2016). The increasing amount of synthetic, LLM-generated data on the internet introduces a precarious feedback loop: LLMs trained on datasets containing synthetic data will themselves generate data that will be used to train future models. Shumailov et al. (2024) demonstrated that this process, known as **recursive training**, can have detrimental effects, causing models to progressively lose information about the true underlying distributions they are intended to approximate. This results in a gradual change in generated distributions, often accompanied by a reduction in variance. While this has sometimes been described as *model collapse*, we refer to this mismatch between the true and the generated distribution as *distribution shift*, and limit the term collapse to refer to *detrimental* distribution shifts, such as losses in quality or diversity. Such detrimental effects (Guo et al., 2024) as well as bias amplification (Wang et al., 2024a) have been reported in previous work. Their potential societal consequences make it imperative to better understand the dynamics of recursive training.

Internet data varies along a wide range of properties. For instance, certain domains may be associated with a higher ratio of synthetic-to-human data (e.g. GitHub), others with lower quality data (e.g. Reddit), and some with lower diversity data (e.g. specialized forums). If those properties affects the outcome of recursive fine-tuning, we may expect different types of shifts on different parts of the internet.

Given the early stage of research in this field, existing studies often ignore this diversity, relying on simplifying assumptions and focusing on abstracted settings. To the best of our knowledge, how different data properties influence distribution shifts remains largely unexplored, aside from studies examining the ratio of human to synthetic data (Bertrand et al., 2023; Bohacek and Farid, 2023; Kazdan et al., 2024; Martínez et al., 2023b; Zhang et al., 2024b). Filing this gap is therefore crucial to draw a more nuanced and detailed picture of the consequences of recursive fine-tuning.

In this paper, we adopt the iterative chain paradigm used in previous studies (Shumailov et al., 2024; Gerstgrasser et al., 2024a). Our experimen-

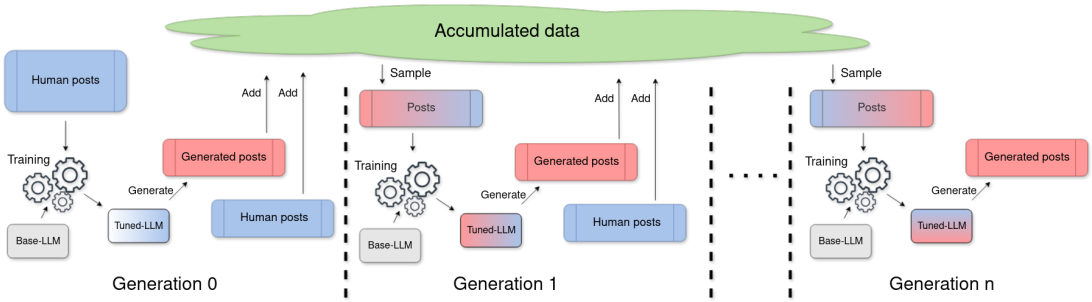

Figure 1: **Iterative chain** In each generation, a fresh base model is fine-tuned on texts sampled from the Accumulated data pool (except generation 0, where it's trained only on human posts). The model generates posts, which are added to the pool alongside some newly sampled human posts.

tal setup is shown in Figure 1. The process begins with fine-tuning a base LLM on a selection of human data (e.g. Reddit posts). This model generates data, which are added to the pool of Accumulated data together with newly sampled human data. In each subsequent generation, a new base model is fine-tuned on data sampled from the Accumulated data pool. This model, in turn, generates new data, which are again added to the pool along with fresh human data. This pipeline allows us to study how generated data evolve across successive generations, with a particular focus on the distribution shift from the first to the final generation.

In our experiments, we study how various properties of human data (e.g. quality, diversity, bias) influence the dynamics of distribution shifts in recursive training chains. We use five datasets spanning three domains (Twitter, Reddit and Wikipedia). First, we confirm that the choice of the dataset greatly influences the consequences of iterative fine-tuning (Section 4.2): while some datasets exhibit sharp decreases in diversity and quality, others are more robust to such shifts. Our second set of experiments (Sections 4.3 and 4.4) aims to uncover specifically which properties of training data mitigate or amplify distribution shifts. We run iterative chain experiments on 800 clusters created from four different datasets, and conduct a series of regression analysis mapping various data properties to the degradation in the quality and diversity of generated texts. We find that lexical diversity is associated with greater degradation, while semantic diversity has the opposite effect. Furthermore, we observe that these influences are highly modular, with generated content being mostly influenced by human data properties from the same domain. This suggests that different internet domains might undergo distinct and relatively independent distribution shifts regardless of models being trained on a mixture of domains. Finally, our last set of experiments focuses on the evolution of political bias. The results indicate that the type of shift observed (bias reduction, amplification or inversion) depends on the political lean of the the human data. This empirically confirms that properties of human data play an important in shaping the outcome of recursive training.

The code for reproducing the simulations, analyses and figures is available on our GitHub[1].

The main contributions of this work are:

- We propose and experimentally confirm the hypothesis that different training datasets lead to different distribution shift dynamics, motivating an investigation on the underlying causes.
- Through an extensive set of experiments (four datasets over three domains), we outline several data properties as influencing distribution shift dynamics.
- We reveal that these influences are highly modular, with generated content being mostly influenced by human data properties from the same domain.
- We find that distribution shifts also occur in terms of political lean, and that the type of shift (bias amplification, reduction or inversion) depends on the political lean of the human data.

2 Related Work

Recursive fine-tuning and model collapse A rapidly growing body of literature has focused

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on the consequences of recursively training generative models on synthetic data (Schaeffer et al., 2025). The phrase "model collapse", coined in Shumailov et al. (2024), refers to the progressive degradation of models induced by this feedback loop. This phenomenon has been studied both theoretically (Dohmatob et al., 2024a,b; Bertrand et al., 2023; Alemohammad et al., 2023) and empirically, on both generative image models (Martínez et al., 2023b,a; Bohacek and Farid, 2023; Hataya et al., 2022; Alemohammad et al., 2023) and language models (Zhang et al., 2024b; Guo et al., 2023; Kazdan et al., 2024; Briesch et al., 2023; Gerstgrasser et al., 2024b). Theoretical results have provided valuable insights, for instance showing how it is characterized by the disappearance of distribution tails (Dohmatob et al., 2024a,b; Shumailov et al., 2024). Empirical studies have allowed to establish several properties of this phenomenon, such as the role of synthetic-to-real-data ratio (Briesch et al., 2023; Hataya et al., 2022) or strategies for mitigating collapse (Gerstgrasser et al., 2024b; Kazdan et al., 2024; Zhang et al., 2024b). Recently, Wang et al. (2024b) showed that recursive LLM fine-tuning can lead to bias amplification. These works do not systematically evaluate how the properties of the human dataset used in their experiments affect their conclusions. Here, we extend this literature by investigating how those properties impact the outcome of recursive training.

Cultural dynamics in artificial agents The motivation for this research area stems from the observation that human-made technologies have transitioned from passive mediators of cultural evolution (e.g., the printing press) to active generators of cultural content. This shift has been described as the emergence of machine culture - culture mediated or generated by machines (Brinkmann et al., 2023). Understanding the dynamics that shape the evolution of machine-generated content over time is therefore crucial. This has led researchers to study cultural dynamics in populations of reinforcement learning agents (Cook et al., 2024; Schmitt et al., 2018; Team et al., 2021; Prystawski et al., 2023; Nisioti et al., 2022) and of LLMs (Perez et al., 2024a,b; Nisioti et al., 2024; Vallinder and Hughes, 2024; Burton et al., 2024). Our work extends this literature by examining the factors that modulate the evolution of LLM-generated content.

3 Methods

3.1 The iterative chain paradigm

We use the iterative chain paradigm inspired by Shumailov et al. (2024); Gerstgrasser et al. (2024a). Our experimental design is shown in Figure 1. First, a base LLM is fine-tuned on 8000 samples from a human dataset (e.g. Wikipedia articles or Reddit posts). This model generates $4000 * r$ posts, where r is the synthetic-data ratio. Those posts are added to the pool of accumulated data together with $4000 * (1 - r)$ newly sampled human posts. In all subsequent generations, a new base model is fine-tuned on 4000 posts sampled from the accumulated data pool, and $4000 * r$ posts generated by this model are added to the accumulated data pool together with $4000 * (1 - r)$ newly sampled human data. In each generation, a new base model is sampled from four possible options: `LLaMa-3.2-1B` (Dubey et al., 2024) (LLAMA 3.1 Community), `Qwen2.5-1.5B` (Team, 2024b) (Apache), `SmolLM-1.7B` (Allal et al., 2024) (Apache), `Falcon3-1B-Base` (Team, 2024a)), and fine-tuned using LoRA (Hu et al., 2021) (see appendix B.5 for details). Five seeds were used in all experiments. This pipeline enables us to study the evolution of generated data over generations, and most notably, the difference between data generated in the first and last generations.

3.2 Datasets

We conducted our experiments on five datasets: two consisting of Twitter posts, two of Reddit posts, and one of Wikipedia paragraphs. These platforms were chosen because they are likely to be increasingly populated with AI-generated content, often indistinguishable from human-written text. Additionally, they are frequently scraped to construct training dataset for language models. Finally, they cover a diverse range of topics and language styles - an essential requirement for investigating the effects of data properties on recursive training dynamics. Refer to Appendix B.2 for details.

3.3 Metrics

In this work, we study distribution shift dynamics in terms of quality, semantic diversity and political lean. Irrespective of how much content a model outputs (which varies with the synthetic-data ratio), we always evaluate those metrics on a sample of 250 generated texts.